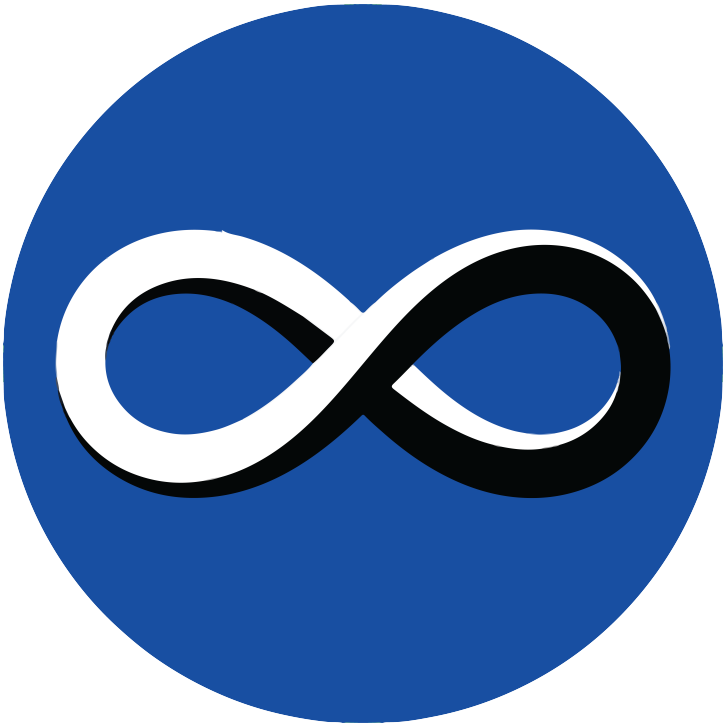
Deputy Coordinator Application

2024-25



MATHEMATICS CLUB



IIT MADRAS

v3.0

Instructions

General

- Provide descriptive answers for all questions. *Explain your thought process well.* Partial points will be awarded for partial solutions.
- You are free to use the internet and any other resources for solving the application. You are **NOT** allowed to collaborate with others. Please cite whatever reference(s) you use.
- The goal of the application is for you to learn as much as possible. Resources have been provided next to questions wherever necessary. All the best and have fun!
- A helpbook has been provided here to make your apping easy on the technical parts. You may find resources relevant to the application on our website.

Application Format

- Your solutions can be manually written, in digital notes, documents or in Latex. Upload all your scanned images, descriptive answers, and others in a single PDF file and submit the final document before the deadline. **There is no weightage to the mode of your submission.** Spend more time on solving the app and not on beautifying it!
- Keep your answers brief and elegant. Explain your thoughts concisely. Ensure to not exceed the word limit indicated next to the question.
- The PDF should be named `<Firstname>_<ROLLNUMBER(IN CAPS)>.pdf`
  (For example: `Atreya_EE22B004.pdf`).

Submission

- The final PDF can be submitted here.
- The deadline for submission is 22nd October, 2024, 23:59
- Join this WhatsApp group for further notifications and updates.

Contact Us

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You can find the contacts of the section masters in the helpbook.

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§0 General Questionnaire

1. Why do you want to be part of the Mathematics Club? (150 words)
2. Have you been to Mathematics Club event(s)? Name and describe your experience in the event(s). (under 150 words)
3. What topic do you hate the most in mathematics? Describe its applications in the real world briefly. (100 words)

§1 An Ocean of Change

Calculus. It is what it is. You've known it for at least the past two years and you'll keep on seeing it for the coming years (unlike the pure joy of your childhood which you left behind in third grade).

1. **Complex Integrals!** Evaluate the integral:

$$\int_\gamma \frac{\sin z}{(z-1)(z-3)} dz$$

where the curve γ is a square centred at the origin with its sides parallel to the coordinate axes and with side length 2 traversed twice in the clockwise direction. (Hint: What do you do when you can't deal with multiple fractions in the denominator?)

If you need help with understanding what a complex integral is then click here to take an adventure over to the helpbook. (3 points)

2. **Feynman for ya!** Evaluate the integral (perhaps using the Feynman technique): (4 points)

$$\int_0^\infty \frac{\log(e+x^2)}{1+x^2} dx$$

§2 A Land of Lines and Shapes

Geometry - the art of correct reasoning from incorrectly drawn figures. You may loathe it or love it, it is here to say hi :)

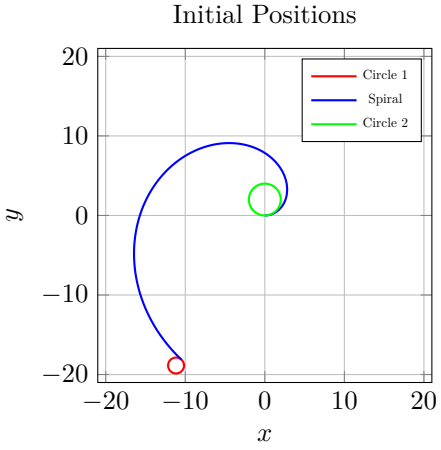
1. **Great Distances:** Derive the shortest path between any two points on an 3-dimensional-sphere when you are constrained to travel only along its surface. Say the first point is $\left(x_1, y_1, \sqrt{1-x_1^2-y_1^2}\right)$ and the second point is $\left(x_2, y_2, \sqrt{1-x_2^2-y_2^2}\right)$ then find this shortest distance. (Hint: Look up great circles) (3 points)

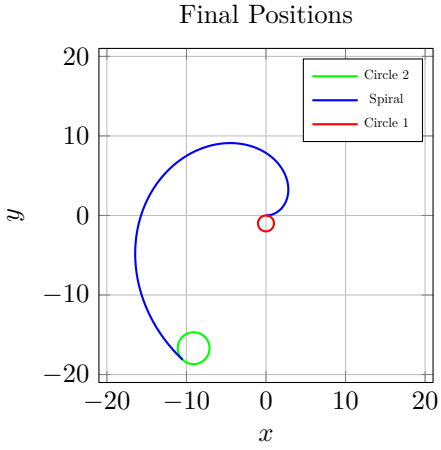
Generalize and prove the above result for n-dimensions if you are up for it! (Bonus)

2. **Rolling rollers:** $Circle_1$ rolls on the outer surface and $Circle_2$ rolls on the inner surface. Find the ratio of rotations (about their centres) of $Circle_1$: $Circle_2$

- Radius of $Circle_1$ $= 1$
- Radius of $Circle_2$ $= 2$
- Equation of spiral : $r = 5\theta,\ \theta \in \left[0, \frac{4\pi}{3}\right]$

(Click here if you don't know how polar coordinates work) (3 points)



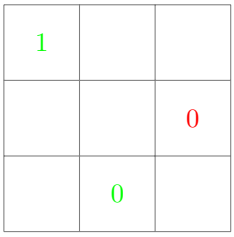


§3 Linear Algebruh

Algebraic symbols are used when you do not know what you are talking about.

§3.1 A truly dependant game

KK and KV are playing a new game involving an $n \times n$ grid of boxes. KV starts by placing a 0 or a 1 anywhere in the grid. KK then follows by again placing a 0 or a 1 in any of the remaining spaces. This continues until there is no spaces left. The goal of the game is determined by the determinant of the resultant $n \times n$ matrix. If the determinant is 0, KK wins, else KV wins. Assuming both KK and KV play optimally, who would win and why?



In the diagram, $n = 3$ is taken as an example and the green numbers are marked by KV and the red numbers are marked by KK. (4 points)

§3.2 Death by Intuition

You learnt about vectors in physics. Now let's play with them in math. Let us define an operator $\Lambda : \mathbb{V} \times \mathbb{V} \to \mathbb{V}$ which is distributive, associative, anti-symmetric and follows: $|\mathbf{e_1}\Lambda\mathbf{e_2}\Lambda.....\Lambda\mathbf{e_n}| = 1$ where $\mathbf{e_i}$ are orthogonal unit vectors in $\mathbb{V}$ i.e. $\mathbf{e_i} \cdot \mathbf{e_j} = 0 \ \ \forall\ i \neq j$. You would have encountered in Cartesian 3D, $(\mathbf{e_1}, \mathbf{e_2}, \mathbf{e_3})$=$(\mathbf{i}, \mathbf{j}, \mathbf{k})$ Now let's try and intuitively understand operator Λ. If you wish to understand what these terms mean click here.

1. Assume two 2D vectors $\mathbf{v_1}$ and $\mathbf{v_2}$. Can you somehow relate $\mathbf{v_1}\Lambda\mathbf{v_2}$ to the cross product? Can you interpret it geometrically? (1 point)

2. Can you extend your previous result to three dimensions? (i.e what do you expect $\mathbf{v_1}\Lambda\mathbf{v_2}\Lambda\mathbf{v_3}$ to be geometrically?) Now prove it. (2 points)

3. Let $\mathbf{v_i} := (v_{i1}, v_{i2}, ...v_{in})$. Can you represent $(\mathbf{v_1}\Lambda\mathbf{v_2}\Lambda.....\Lambda\mathbf{v_n})$ in terms of the components of $\mathbf{v_i}$; $i = 1, 2, 3...n$ without any vector operations in some compact form. (Hint: Think of matrices and the cases you proved above.) Proof is not necessary (but will earn brownie points :)). Intuition is sufficient. (2 points)

4. Using the result in the previous part, can you prove a property about $\dfrac{d}{dt}(\mathbf{v_1}\Lambda\mathbf{v_2}\Lambda.....\Lambda\mathbf{v_n})$ given that $\mathbf{v_k}$ are dependent on t for all k. (1 points)

§4 A Network of Connections

Graphs aren't just for math—they're everywhere! They help us understand how things are connected and interact. A Graph is nothing but like a network of friends. Each friend is a node and the connections between them are represented by edges. Here are a few definitions.

- Vertices (or Nodes): These are the points in a graph. Think of each person as a vertex.
- Edges: These are the lines that connect two vertices. For example an edge shows that two people are friends.
- Degree of a vertex: This tells us how many edges are connected to a vertex. In other words, the degree of a person (vertex) is how many friends they have.

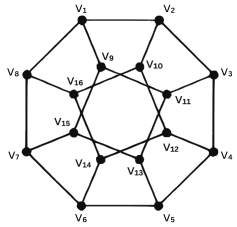
1. **Exploring Degree Sum and Number of Edges:** Now that we know what vertices, edges, and degrees are, let's dig a little deeper! Can you figure out how the sum of all degrees relates to the total number of edges (friendships) in the network? In other words imagine you have a network of friends. Each person (vertex) has a certain number of friends (degree). If we add up the number of friends everyone has, how can we figure out the total number of friendships? (2 points)

2. **Chessboard with Graphs:** We can model the chessboard as a graph in which each square is represented as a node. An edge connects two nodes if a knight can legally move from one square to the other. Your goal is to investigate the characteristics of this graph representing the knight's movement!
   - Is the graph Bipartite? If it is Bipartite, try to give a Bipartition, else explain why. (2 points)
   - In general, try to think of a condition for a graph to be bipartite. (2 points)

3. **Hamiltonian and Eulerian:** A graph is called Hamiltonian if there exists a cycle (called a Hamiltonian cycle) that visits every vertex of the graph exactly once and returns to the starting vertex.

   A graph is called Eulerian if there exists a cycle (called an Eulerian cycle or Eulerian circuit) that traverses every edge of the graph exactly once and returns to the starting vertex.

   Now consider the graph G:



   a) Determine whether the graph G is Hamiltonian (Give a Hamiltonian cycle if it exists). (2 points)

   b) Determine if the graph is Eulerian. What is a necessary condition for any graph to be Eulerian? (2 points)

§5 The Limits of Logic

Logical reasoning forms a very important part of framing mathematical arguments. This is an essential skill that helps you in proving and disproving claims that your fellow math enthusiasts may make (and is also helpful in proving that your friend is an RG). This section is designed to test how well you observe a given problem and how well you are able to put forth your arguments in favour of a claim.

1. **Function foundry:** Let $\mathbb{A} \subset \mathbb{R}_+$, that is $\mathbb{A}$ is a subset of the positive reals. Now $f : \mathbb{A} \to \mathbb{R}$ is a function that satisfies the condition:

$$f(x)f(y) = \frac{1}{x+y} \quad \forall x, y \in \mathbb{A}$$

   Find a function f along with your choice of $\mathbb{A}$ which satisfy the aforementioned condition. (3 points)

2. **Back to basics:** Let $f : \mathbb{R}_+ \to \mathbb{R}$ be given by $f(x) = x - \frac{1}{x}$ for all $x \in \mathbb{R}_+$. Now let

$$g(x) = f\left(f(x)\right) - f\left(f\left(\frac{1}{x}\right)\right)$$

   where g is defined on the most exhaustive domain possible. What kind of function is g? Is g a special kind of function? (2 points)

3. **Proof Theory!**

   a) **The easy one:** In a group of n friends, a person can shake hands with any number of other friends. Prove that there are at least two friends who shook their hands with the same number of people. Start by stating your proof technique and why you chose it. (2 points)

   b) **Do it by Exhaustion:** Due to their ongoing rivalry, Atreya and Prad in their settlement chose to take half the set of coords with them. Each coord from Atreya's team chooses a best friend from Prad's team and vice versa, if two coords choose each other they are considered mutual best friends. Prove that if there are infinitely many coords, the probability that there are no mutual best friends is $\frac{1}{e}$ (3 points)

   c) **Torn realities:** Show that if there cannot exist any continuous map from the disk to its entire boundary then continuous maps of the disk onto itself must have at least one fixed point. If you don't understand what this statement means, here is a short read to help you out! (Bonus)

1. Why do you want to be part of the Mathematics Club? (150 words)

I want to be a part of the Mathematics Club because I am passionate and would like to explore more mathematical concepts. Engaging with seniors in this club will enhance my understanding and appreciation of mathematics. The collaborative environment will also allow to tackle challenging problems and share innovative ideas.

Additionally, I believe that being involved in this club will help me develop critical thinking and problem-solving skills, which are essential in real-world situations. I am eager to learn from others and contribute my unique perspective, fostering a community of support and growth.

Moreover, I see this as an opportunity to discover various applications of mathematics, from theoretical aspects to practical uses in science and technology. Overall, joining the Mathematics Club aligns perfectly with my academic goals and desire to cultivate lasting friendships with like-minded individuals.

2. Have you been to Mathematics Club event(s)? Name and describe your experience in the event(s). (under 150 words)

I have had the opportunity to attend a few events organized by the Mathematics Club, each of which has left a lasting impression.

Mathematics Club and Cybersecurity Club Collaboration: This was my first event with the Mathematics Club, and it provided a highly positive introduction. Participants were paired with a partner, and the event culminated in a test, which I found to be both engaging and challenging.

Integration Bee: This event stands out as one of my favorites. Initially, my partner and I did not expect to advance to the top 32. However, we took the competition more seriously after qualifying and ultimately made it to the top 12 before being eliminated. The experience was immensely rewarding, as it involved solving interesting and complex problems, with discussions on problem-solving techniques held afterward.

Strategy Games and Rivalry: The most recent event I attended, as of October 2024, was Strategy Games and Rivalry. The name itself piqued my interest, and my friend and I decided to participate. We engaged in various games, developing strategies (in the pursuit of chocolates), and found it to be an entirely new and enjoyable experience.

3. What topic do you hate the most in mathematics? Describe its applications in the real world briefly. (100 words)

I hate combinatorics the most in mathematics, as I often find its complexity overwhelming and confusing. However, its applications in the real world are significant. Combinatorics plays a crucial role in fields such as computer science, where it is used in algorithm design and data analysis. It helps in optimizing resource allocation, like determining the most efficient routes for logistics companies. Additionally, combinatorial principles are essential in probability theory, influencing everything from statistical modeling to game theory. Despite my dislike for it, I recognize that combinatorics underpins many practical problems and solutions we encounter daily.

1. **Complex Integrals!** Evaluate the integral:

$$\int_\gamma \frac{\sin z}{(z-1)(z-3)} dz$$

where the curve γ is a square centred at the origin with its sides parallel to the coordinate axes and with side length 4 traversed twice in the clockwise direction. (Hint: What do you do when you can't deal with multiple fractions in the denominator?)

If you need help with understanding what a complex integral is then click here to take an adventure over to the helpbook. (3 points)

Used addendum as reference for this question

$n(8,1), n(8,3) = 2$ (-ve sign due to clockwise rot^n)

$$2 \cdot \sin 3 = \frac{-1}{2\pi i} \int_\gamma \frac{\sin z}{z-3} dz \quad — (1)$$

equation from resource provided

$$2 \cdot \sin 1 = \frac{-1}{2\pi i} \int_\gamma \frac{\sin z}{z-1} dz \quad — (2)$$

Subtract (1) and (2)

$$2(\sin 3 - \sin 1) = \frac{-1}{2\pi i} \int_\gamma \sin z \left[\frac{z-1-z+3}{(z-1)(z-3)}\right] dz$$

$$\Rightarrow 2(\sin 3 - \sin i) = \frac{-1}{\pi i} \int_\gamma \frac{\sin z \, dz}{(z-1)(z-3)}$$

$$\Rightarrow \boxed{\int_\gamma \frac{\sin z}{(z-1)(z-3)} dz = -2\pi i (\sin 3 - \sin 1)}$$

2. **Feynman for ya!** Evaluate the integral (perhaps using the Feynman technique): (4 points)

$$\int_0^\infty \frac{\log(e+x^2)}{1+x^2}dx$$

$$\int_0^\infty \frac{\log(e+x^2)}{1+x^2}\,dx$$

$$I(c) = \int_0^\infty \frac{\log(e+cx^2)}{1+x^2}\,dx$$

$$I'(c) = \int_0^\infty \frac{x^2}{(1+x^2)(e+cx^2)}\,dx = \frac{1}{c-e}\int_0^\infty \left(\frac{1}{1+x^2} - \frac{e/c}{e/c+x^2}\right)dx$$

$$= \frac{1}{c-e}\left[\frac{\pi}{2} - e/c\int_0^\infty \frac{dx}{(\sqrt{e/c})^2+x^2}\right] = \frac{1}{c-e}\left[\frac{\pi}{2} - \sqrt{e/c}\,\frac{\pi}{2}\right]$$

$$\frac{dI}{dc} = \frac{\pi}{2}\,\frac{1}{\sqrt{c}(\sqrt{c}+\sqrt{e})}$$

$$I(0) = \pi/2 \Rightarrow I(c) = \frac{\pi}{2}\int_0^c \frac{dc}{(\sqrt{c}+\sqrt{e})\sqrt{c}} = \pi\log\left(\frac{\sqrt{c}+\sqrt{e}}{\sqrt{e}}\right) + K$$

$K = +\pi/2$

$$I(c) = \pi\log(\sqrt{c/e}+1) + \pi/2$$

$$I(1) = \pi\log(1/\sqrt{e}+1) + \pi\ln\sqrt{e}$$

$$\boxed{\int_0^\infty \frac{\log(e+x^2)}{1+x^2}\,dx = \pi\log(1+\sqrt{e})}$$

Later I realized another way to the same problem with Feynman!

$$I(C) = \int_0^\infty \frac{\log(C+x^2)}{1+x^2}\,dx$$

$$I'(C) = \int_0^\infty \frac{dx}{(C+x^2)(1+x^2)} = \frac{1}{C-1}\int_0^\infty \left(\frac{1}{1+x^2} - \frac{1}{C+x^2}\right)dx$$

$$= \frac{1}{C-1}\left[\frac{\pi}{2} - \frac{\pi}{2\sqrt{C}}\right] = \frac{\pi}{2}\,\frac{\sqrt{C}-1}{(C-1)\sqrt{C}} = \frac{\pi}{2\sqrt{C}(\sqrt{C}-1)}$$

$$I(C) = \int \frac{\pi\,dC}{2\sqrt{C}(\sqrt{C}+1)} = \pi\log(\sqrt{C}+1) + K$$

$$I(0) = \int_0^\infty \frac{\log x^2}{1+x^2}\,dx = 2\int_0^\infty \frac{\log x}{1+x^2}\,dx$$

put $x = \tan\theta \Rightarrow dx = \sec^2\theta\,d\theta$

$$I(0) = 2\int_0^{\pi/2} \log(\tan\theta)\,d\theta = 2\int_0^{\pi/2} \log(\cot\theta)\,d\theta$$

(King's rule)

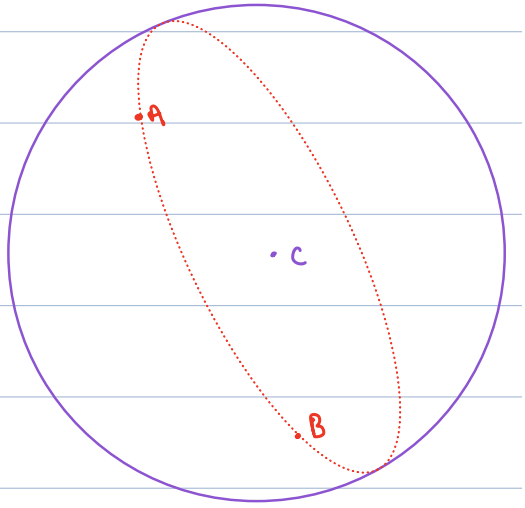
$$2I(0) = 2\int_0^{\pi/2} \log 1\,d\theta = 0 \Rightarrow I(0) = 0 \Rightarrow K = 0$$

$$I(C) = \pi\log(\sqrt{C}+1) \Rightarrow \boxed{I(e) = \pi\log(\sqrt{e}+1)}$$

1. **Great Distances:** Derive the shortest path between any two points on an 3-dimensional-sphere when you are constrained to travel only along its surface. Say the first point is $\left(x_1, y_1, \sqrt{1-x_1^2-y_1^2}\right)$ and the second point is $\left(x_2, y_2, \sqrt{1-x_2^2-y_2^2}\right)$ then find this shortest distance. (Hint: Look up great circles) (3 points)

Used addendum for great circle part and went through some publications for reference

The shortest distance between A and B (2 points on a sphere) (we also travel on the sphere) is an arc of the great circle (circle that pertains to a point and contains the center of sphere)



Using spherical polar coordinates,

$z = r\cos\theta \quad x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$\vec{ds} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}$

$|\vec{ds}| = \sqrt{r^2 d\theta^2 + r^2\sin^2\theta\, d\phi^2}$

let $s = s(t)$; $\theta = \theta(t)$; $\phi = \phi(t)$ $\quad t \in [0,1]$ (A at 0, B at 1)

$$S = \int_A^B |\vec{ds}| = \int_A^B \sqrt{d\theta^2 + \sin^2\theta\, d\phi^2} \qquad (r=1) \quad \text{——} (2)$$

$$L = \int_0^1 \sqrt{(d\theta/dt)^2 + \sin^2\theta\,(d\phi/dt)^2}\, dt \longrightarrow \text{to be minimized}$$

$\sqrt{(d\theta/dt)^2 + \sin^2\theta\,(d\phi/dt)^2} = f$

By Euler Lagrangian Method

$$\frac{dF}{d\theta} - \frac{d}{dt}\left(\frac{dF}{d\theta'}\right) = 0 \qquad \frac{dF}{d\phi} - \frac{d}{dt}\left(\frac{dF}{d\phi'}\right) = 0$$

$$\frac{\sin\theta\cos\theta(\phi')^2}{\sqrt{(\theta')^2+\sin^2\theta(\phi')^2}} - \frac{d}{dt}\left(\frac{\theta'}{\sqrt{(\theta')^2+\sin^2\theta(\phi')^2}}\right)=0 \quad -(2)$$

$$-\frac{d}{dt}\left(\frac{\sin^2\theta(\phi')}{\sqrt{(\theta')^2+\sin^2\theta(\phi')^2}}\right)=0 \quad -(3)$$

From (3) $\Rightarrow \dfrac{\sin^2\theta(\phi')}{\sqrt{(\theta')^2+\sin^2\theta(\phi')^2}} = k$

$$\Rightarrow \sin^4\theta(\phi')^2 = k^2[(\theta')^2+\sin^2\theta(\phi')^2]$$

$$\Rightarrow \sin^2\theta[\sin^2\theta-k^2](\phi')^2 = k^2(\theta')^2$$

Introducing new variable w such that

$$\sin\theta(\phi') = \cos w(s')$$

$$(\theta') = \sin w(s')$$

w represents angle between tangent vector to the curve and line of latitude (constant θ line)

$$[\sin^2\theta-k^2]\cos^2 w = k^2\sin^2 w$$

$\Rightarrow \sin^2\theta\cos^2 w = k^2$ $\longrightarrow$ this equation is independent of s. which means w is constant along the path. (onstant w describes a

plane through origin. The intersection of this plane with the sphere is a great circle.

Now to find the minimum value of the distance

$$\cos\theta = x_1x_2 + y_1y_2 + z_1z_2 \Rightarrow \theta = \cos^{-1}(x_1x_2 + y_1y_2 + z_1z_2)$$

$$S = r\theta = 1 \times \theta = \theta = \cos^{-1}(x_1x_2 + y_1y_2 + z_1z_2)$$

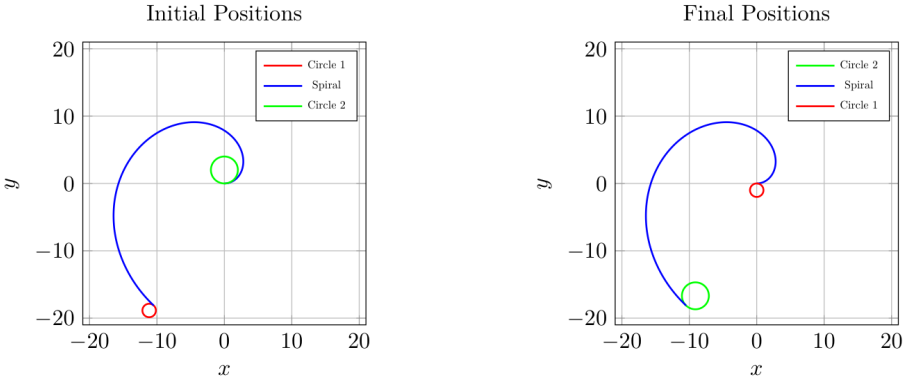
$$\boxed{S = \cos^{-1}(x_1x_2 + y_1y_2 + z_1z_2)}$$

In n-dimensional system, let's assume a set of n points at a constant distance (r) from the origin. Assume the coordines as $A(a_1, a_2, \ldots, a_n)$ & $B(b_1, b_2, \ldots, b_n)$. Then the shortest distance from A to B along the surface is $r\cos^{-1}\left(\frac{a_1b_1 + a_2b_2 + \ldots a_nb_n}{r^2}\right)$

2. **Rolling rollers:** $Circle_1$ rolls on the outer surface and $Circle_2$ rolls on the inner surface. Find the ratio of rotations (about their centres) of $Circle_1$: $Circle_2$

main idea is to find the arc length traversed by the center of the circles.

- Radius of $Circle_1$ $= 1$
- Radius of $Circle_2$ $= 2$
- Equation of spiral : $r = 5\theta,\ \theta \in \left[0, \frac{4\pi}{3}\right]$
  (Click here if you don't know how polar coordinates work) (3 points)



Arc length $= \int_{\alpha}^{\beta} \sqrt{(\dot{x})^2 + (\dot{y})^2}\, dt$ $\quad \dot{x} = \frac{dx}{dt}$ $\quad \dot{y} = \frac{dy}{dt}$

$r = f(\theta) \Rightarrow x = f(\theta)\cos\theta,\ y = f(\theta)\sin\theta$

Arc length $= \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$

Circle 1: $r = 5\theta + \underbrace{1}_{\text{center}}$ $\quad \frac{dr}{d\theta} = 5$

distance travelled by center of Circle 1:

$$\int_0^{4\pi/3} \sqrt{25 + (5\theta + 1)^2}\, d\theta = I_1$$

Circle 1: $r = 5\theta - \underbrace{2}_{\text{center}}$ $\quad \frac{dr}{d\theta} = 5$

distance travelled by center of Circle 1:

$$\int_0^{4\pi/3} \sqrt{25+(5\theta-2)^2}\, d\theta = I_2$$

Number of revolutions of circle $1 = \frac{I_1}{2\pi}$

Number of revolutions of circle $2 = \frac{I_2}{4\pi}$

Required ratio $= \frac{I_1}{I_2} \times 2 \approx 2.4323$

§3.1 A truly dependant game

KK and KV are playing a new game involving an $n \times n$ grid of boxes. KV starts by placing a 0 or a 1 anywhere in the grid. KK then follows by again placing a 0 or a 1 in any of the remaining spaces. This continues until there is no spaces left. The goal of the game is determined by the determinant of the resultant $n \times n$ matrix. If the determinant is 0, KK wins, else KV wins. Assuming both KK and KV play optimally, who would win and why?

1		
		0
	0	

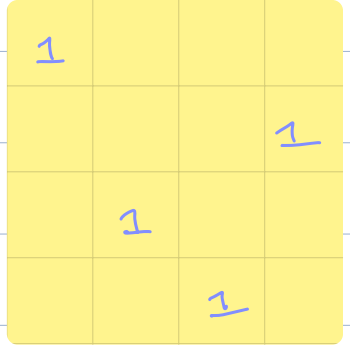
In the diagram, $n = 3$ is taken as an example and the green numbers are marked by KV and the red numbers are marked by KK. (4 points)

If n=1: KV puts a 1 and wins (KV)

If n=2: KK is more likely to win as whatever KV does KK can force the determinant to zero (KK)

If n≥3: KV starts first. whatever efforts KV put KK can just somehow make the determinant zero. For det to be non zero KV will have to put 1's in such a way that every row has 1 in different positions and same with every column. But KK can ruin KV's winning chance by putting 0's at such positions and make the det zero. (KK)

So I feel KK is more likely to win the game if both play optimally.



KV should have some structure like this to win but KK can easily defend it!

You learnt about vectors in physics. Now let's play with them in math. Let us define an operator $\Lambda : \mathbb{V} \times \mathbb{V} \to \mathbb{V}$ which is distributive, associative, anti-symmetric and follows: $|\mathbf{e_1}\Lambda\mathbf{e_2}\Lambda.....\Lambda\mathbf{e_n}| = 1$ where $\mathbf{e_i}$ are orthogonal unit vectors in $\mathbb{V}$ i.e. $\mathbf{e_i} \cdot \mathbf{e_j} = 0 \ \ \forall\, i \neq j$. You would have encountered in Cartesian 3D, $(\mathbf{e_1}, \mathbf{e_2}, \mathbf{e_3})$=$(\mathbf{i}, \mathbf{j}, \mathbf{k})$ Now let's try and intuitively understand operator Λ. If you wish to understand what these terms mean click here.

orientation based

1. Assume two 2D vectors $\mathbf{v_1}$ and $\mathbf{v_2}$. Can you somehow relate $\mathbf{v_1}\Lambda\mathbf{v_2}$ to the cross product? Can you interpret it geometrically? (1 point)
2. Can you extend your previous result to three dimensions? (i.e what do you expect $\mathbf{v_1}\Lambda\mathbf{v_2}\Lambda\mathbf{v_3}$ to be geometrically?) Now prove it. (2 points)
3. Let $\mathbf{v_i} := (v_{i1}, v_{i2}, ...v_{in})$. Can you represent $(\mathbf{v_1}\Lambda\mathbf{v_2}\Lambda.....\Lambda\mathbf{v_n})$ in terms of the components of $\mathbf{v_i}$; $i = 1, 2, 3...n$ without any vector operations in some compact form. (Hint: Think of matrices and the cases you proved above.) Proof is not necessary (but will earn brownie points :)). Intuition is sufficient. (2 points)
4. Using the result in the previous part, can you prove a property about $\frac{d}{dt}(\mathbf{v_1}\Lambda\mathbf{v_2}\Lambda.....\Lambda\mathbf{v_n})$ given that $\mathbf{v_k}$ are dependent on t for all k. (1 points)

Used addendum for going through definitions and properties of the operator

1. $v_1 \wedge v_2$ gives a quantity (plane) with area as the parallelogram defined by the 2 vectors.
$v_1 \times v_2$ gives a vector perpendicular to the plane of v_1 and v_2, and area same as the parallelogram defined by the 2 vectors.

$\Rightarrow |v_1 \wedge v_2| = |v_1 \times v_2|$ where $||$ denotes magnitude

2. I think it is the volume of parallelepiped formed by v_1, v_2, v_3. $|v_1 \wedge v_2 \wedge v_3| = |v_1 . (v_2 \times v_3)|$.
If we resolve v_1, v_2, v_3 into components.

$$v_1 = (v_1)_x \hat{e}_1 + (v_1)_y \hat{e}_2 + (v_1)_z \hat{e}_3$$
$$v_2 = (v_2)_x \hat{e}_1 + (v_2)_y \hat{e}_2 + (v_2)_z \hat{e}_3$$
$$v_3 = (v_3)_x \hat{e}_1 + (v_3)_y \hat{e}_2 + (v_3)_z \hat{e}_3$$
$$\hat{e}_1 \wedge \hat{e}_1 = \hat{e}_2 \wedge \hat{e}_2 = \hat{e}_3 \wedge \hat{e}_3 = 0$$

$$v_1 \wedge v_2 \wedge v_3 = [(v_1)_x (v_2)_y (v_3)_z - (v_1)_x (v_2)_z (v_3)_y + (v_1)_y (v_2)_z (v_3)_x$$

By distributive law

$$- (v_1)_y (v_2)_x (v_3)_z + (v_1)_z (v_2)_x (v_3)_y - (v_1)_z (v_2)_y (v_3)_x] \hat{e}_1 \wedge \hat{e}_2 \wedge \hat{e}_3$$

$$\Rightarrow v_1 \wedge v_2 \wedge v_3 = \underbrace{\det(v_1, v_2, v_3)}_{\text{represents volume of parallelepiped magnitude}} \hat{e}_1 \wedge \hat{e}_2 \wedge \hat{e}_3$$

3. We can generalize from above that

$$v_1 \wedge v_2 \wedge v_3 \ldots \wedge v_n = \det(v_1, v_2, v_3, \ldots, v_n)(\hat{e}_1 \wedge \hat{e}_2 \wedge \hat{e}_3 \ldots \wedge \hat{e}_n)$$

$\det(v_1, v_2, \ldots)$ means det of $n \times n$ matrix with entries as coefficients / components of $v_1, v_2, \ldots$ along $\hat{e}_1, \hat{e}_2, \ldots$.

4. $$\frac{d}{dt}(v_1 \wedge v_2 \wedge \ldots \wedge v_n) = \frac{d}{dt}[\det(v_1, v_2, \ldots, v_n)\, \hat{e}_1 \wedge \hat{e}_2 \ldots \wedge \hat{e}_n]$$

$$= \left[\det\left(\frac{dv_1}{dt}, v_2, \ldots v_n\right) + \det\left(v_1, \frac{dv_2}{dt}, \ldots v_n\right) + \ldots \right.$$

$$\left. \ldots + \det\left(v_1, v_2, \ldots, \frac{dv_n}{dt}\right)\right](\hat{e}_1 \wedge \hat{e}_2 \wedge \ldots \wedge \hat{e}_n)$$

$$= \left(\frac{dv_1}{dt} \wedge v_2 \wedge \ldots v_n\right) + \left(v_1 \wedge \frac{dv_2}{dt} \wedge \ldots v_n\right) + \ldots + \left(v_1 \wedge v_2 \wedge \ldots \frac{dv_n}{dt}\right)$$

Some how matches product rule in differentiation.

1. **Exploring Degree Sum and Number of Edges:** Now that we know what vertices, edges, and degrees are, let's dig a little deeper! Can you figure out how the sum of all degrees relates to the total number of edges (friendships) in the network? In other words imagine you have a network of friends. Each person (vertex) has a certain number of friends (degree). If we add up the number of friends everyone has, how can we figure out the total number of friendships? (2 points)

Each edge connects 2 vertices. So if we count the degree of each vertex, we are basically counting each edge twice. Therefore

Sum of all degrees = 2x total number of edges

2. **Chessboard with Graphs:** We can model the chessboard as a graph in which each square is represented as a node. An edge connects two nodes if a knight can legally move from one square to the other. Your goal is to investigate the characteristics of this graph representing the knight's movement!
   - Is the graph Bipartite? If it is Bipartite, try to give a Bipartition, else explain why. (2 points)
   - In general, try to think of a condition for a graph to be bipartite. (2 points)

Used google for bipartite definition and understanding

If a knight is in a white square, then after it makes a move it must be in a black square (and vice versa). The graph can indeed be divided into 2 sets i.e set with black color squares and set with white color squares. And since none of the move can be from one color to the same color (adjacent). Hence the graph representing the knight's movement is Bipartite. The 2 sets are:

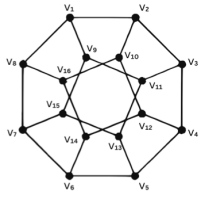
- Set of all white squares.
- Set of all black squares.

While going through the definition of Bipartite graphs. I came through this term odd-length cycles. I came to an understanding that an odd-length cycle is a cycle that has an odd number of edges/vertices. So bipartite graph shouldn't have odd-length cycles. If an odd-length cycle exists in a graph, and you try to alternate between the 2 sets along the edges, you'd return to the vertex in the same set which contradicts the definition of a Bipartite graph. So I feel that a Bipartite graph shouldn't have any odd-length cycle, in contrast it can have an even length cycle.

3. **Hamiltonian and Eulerian:** A graph is called Hamiltonian if there exists a cycle (called a Hamiltonian cycle) that visits every vertex of the graph exactly once and returns to the starting vertex.

   A graph is called Eulerian if there exists a ~~cycle~~ circuit (called an ~~Eulerian cycle or~~ Eulerian circuit) that traverses every edge of the graph exactly once and returns to the starting vertex.

   Now consider the graph G:



   a) Determine whether the graph G is Hamiltonian (Give a Hamiltonian cycle if it exists). (2 points)

   b) Determine if the graph is Eulerian. What is a necessary condition for any graph to be Eulerian? (2 points)

a) Graph G is Hamiltonian.

$V_1 \to V_2 \to V_3 \to V_4 \to V_5 \to V_6 \to V_7 \to V_8 \to V_{16} \to V_{15} \to V_{14}$

$V_1 \leftarrow V_9 \leftarrow V_{10} \leftarrow V_{11} \leftarrow V_{12} \leftarrow V_{13}$

b) Graph G is not Eulerian.

For a graph to be Eulerian, each vertice needs to have an even number of edges connecting it. If it isn't even then we'd have to enter and leave that vertex uneven number of times. Hence, a graph can be Eulerian if all its vertices have an even degree.

1. **Function foundry:** Let $\mathbb{A} \subset \mathbb{R}_+$, that is $\mathbb{A}$ is a subset of the positive reals. Now $f : \mathbb{A} \to \mathbb{R}$ is a function that satisfies the condition:

$$f(x)f(y) = \frac{1}{x+y} \quad \forall x, y \in \mathbb{A}$$

Find a function f along with your choice of $\mathbb{A}$ which satisfy the aforementioned condition. (3 points)

I can assume $A = \{2\}$ and $f(x) = \frac{1}{x}$

I can assume A such that $x+y=c$ $(x,y>0)$ then $f(x) = \frac{e^x}{\sqrt{c}e^{\sqrt{c}}}$ or $f(x) = \pm\frac{1}{\sqrt{c}}$. Even a complex function like $f(x) = \frac{1}{\sqrt{c}} e^{i(\pi x/c)}$ can be defined. All these I've found by assuming $f(x)$ as something and then make an approach. I don't know a proper method to solve this.

2. **Back to basics:** Let $f : \mathbb{R}_+ \to \mathbb{R}$ be given by $f(x) = x - \frac{1}{x}$ for all $x \in \mathbb{R}_+$. Now let

$$g(x) = f\left(f(x)\right) - f\left(f\left(\frac{1}{x}\right)\right)$$

where g is defined on the most exhaustive domain possible. What kind of function is g? Is g a special kind of function? (2 points)

Since domain of f is only positive reals, $f(x)$ and $f(1/x)$ must be positive.

$x - 1/x > 0$ and $1/x - x > 0$

No such x satisfies both conditions at once.

$\Rightarrow$ Domain of $g(x)$ is null ($\emptyset$). I don't think we can comment anything else about g.

3. **Proof Theory!**

a) **The easy one:** In a group of n friends, a person can shake hands with any number of other friends. Prove that there are at least two friends who shook their hands with the same number of people. Start by stating your proof technique and why you chose it. (2 points)

b) **Do it by Exhaustion:** Due to their ongoing rivalry, Atreya and Prad in their settlement chose to take half the set of coords with them. Each coord from Atreya's team chooses a best friend from Prad's team and vice versa, if two coords choose each other they are considered mutual best friends. Prove that if there are infinitely many coords, the probability that there are no mutual best friends is $\frac{1}{e}$ (3 points)

a) I am going to use proof by contradiction. I'm chosing this method as I've done previous questions like this before.

Let's say there are n people and no two people with same number of handshakes. So that means each guy has a different number of handshakes. The maximum number of handshakes a person can do is n-1 (All but him), and minimum is 0 (no one). There are n possible handshake's (0 to n-1). So each person would have shook hands with a distinct number from 0 to n-1. n people got n different 'handshake values'. Now since the person with n-1 handshakes would have shaken hands with everyone but him, there's no way there is a person with 0 handshakes. But there is one

person according to our assumption. Hence our assumption is wrong. Therefore in a group of n people, there exists Atleast 2 people with same number of handshakes.

Used Wikipedia for derangement concept

b) This sounds similar to number of derangements.

$D_n = n! \sum_{i=0}^{n} \frac{(-1)^i}{i!}$ [permute n coordinates E none of them is paired with corresponing coordinate from next team]

$\Rightarrow$ Probability of no mutual best friends

$$P = \frac{D_n}{n!} = \sum_{i=0}^{n} \frac{(-1)^i}{i!}$$

As $n \to \infty$; $P = \sum_{i=0}^{\infty} \frac{(-1)^i}{i!} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots$

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$ (from power series)

we can see that $e^{-1} = 1 - 1 + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \cdots$

$\Rightarrow \boxed{P = \frac{1}{e}}$

c) **Torn realities:** Show that if there cannot exist any continuous map from the disk to its entire boundary then continuous maps of the disk onto itself must have at least one fixed point. If you don't understand what this statement means, here is a short read to help you out! (Bonus)

Used addendum for reference and CHATGPT to understand the question

Suppose there is no fixed point for the continuous map. means $f(x)$ can't be x $\forall x \in D$. Now build a map from points inside the disc to points on boundary δD. Consider a line joining $f(x)$ and x $\forall x \in D$. We want to push x in D along the line towards the boundary δD. This line would intersect the boundary at some point. Now for each point x in D, map it to that point where the line meets the boundary. This now maps the whole disc onto its boundary. But this a contradiction as the addendum says there cannot exist a continuous map from the disc to it boundary. Hence our assumption that there is no fixed points is wrong. Therefore the continuous map must have atleast one fixed point.

(a) If $f : D \to D$ is a continuous map, then $f(x) = x$ for atleast one point $x \in D$. Think of this as there being at least one immovable point in any continuous map from a disk onto itself. For example the rotation of the disk about its centre has an immovable point - the centre itself.

(b) There cannot exist an onto continuous map $g : D \to \partial D$. That is there does not exist any continuous map from the disk to the entirety of its boundary. You cannot stretch the interior of the elastic cloth to the exterior without creating holes in it and ripping it apart!